

### Random Forest Classification

| precision    | recall | f1-score | support |      |
|--------------|--------|----------|---------|------|
|              |        |          |         |      |
|              | 0      | 0.93     | 0.92    | 0.92 |
|              | 1      | 0.86     | 0.88    | 0.87 |
|              |        |          |         |      |
| accuracy     |        |          |         | 0.90 |
| macro avg    |        | 0.89     | 0.90    | 0.90 |
| weighted avg |        | 0.90     | 0.90    | 0.90 |

### Decision Tree classification

| precision    | recall | f1-score | support |      |
|--------------|--------|----------|---------|------|
|              |        |          |         |      |
|              | 0      | 0.88     | 0.92    | 0.90 |
|              | 1      | 0.84     | 0.78    | 0.81 |
|              |        |          |         |      |
| accuracy     |        |          |         | 0.87 |
| macro avg    |        | 0.86     | 0.85    | 0.85 |
| weighted avg |        | 0.86     | 0.87    | 0.86 |

### Support Vector Machine Classification

| precision    | recall | f1-score | support |      |
|--------------|--------|----------|---------|------|
|              |        |          |         |      |
|              | 0      | 0.76     | 0.96    | 0.85 |
|              | 1      | 0.88     | 0.47    | 0.61 |
|              |        |          |         |      |
| accuracy     |        |          |         | 0.78 |
| macro avg    |        | 0.82     | 0.72    | 0.73 |
| weighted avg |        | 0.81     | 0.78    | 0.76 |

Out of 3 model, Random Forest classification has least values

### Question from my knowledge

### Question for Random forest classification

#### 1. How many class present in dataset?

Dataset has 2 class :

Class 0 – Not purchased - 85

Class 1 – purchased - 49

#### 2. Does dataset is balanced?

No dataset is imbalanced, because class 0 has 85 samples, class 1 has 49 samples

#### 3. Whats the overall performance of model?

Accuracy – 90%, precision – 0.90, Recall – 0.90, and F1 score = 0.90

#### **4. what the precision?**

class 0: 0.93, out of 85 samples, 93 % were identified as not purchased (irrespective of correct and incorrect)

class 1: 0.86, out of 49 samples, 86 % were identified as purchased irrespective of correct and incorrect)

#### **5. whats the recall value?**

Class 0: 0.92, out of 85, samples, 92 % were correctly identified as not purchased

Class 1: 0.88, out of 49 samples, 88 % were correctly identified as purchased

#### **6. whats the F1-score value?**

Class 0: Excellent balance between Recall and precision (0.92)

Class 1: Good balance between Recall and precision (0.87)

#### **7. whats the overall classification report?**

"The model achieved an overall accuracy of 90%, meaning it correctly classified 90% of the test data. For Class 0 (Not Purchased), the precision is 93%, recall is 92%, and F1-score is 92%, showing excellent performance. For Class 1 (Purchased), the precision is 86%, recall is 88%, and F1-score is 87%, indicating good performance, although the model misses some customers who actually purchased. Overall, the weighted F1-score is 86%, which shows the model performs well across both classes.

### **Question for Decision Tree classification**

#### **1. How many class present in dataset?**

Dataset has 2 class :

Class 0 – Not purchased - 85

Class 1 – purchased - 49

#### **2. Does dataset is balanced?**

No dataset is imbalanced, because class 0 has 85 samples, class 1 has 49 samples

#### **3. Whats the overall performance of model?**

Accuracy – 87% precision – 0.81, Recall – 0.85, and F1 score = 0.86

#### **4. what the precision?**

class 0: 0.88, out of 85 samples, 88 % were identified as not purchased (irrespective of correct and incorrect)

class 1: 0.84, out of 49 samples, 84 % were identified as purchased irrespective of correct and incorrect)

#### **5. whats the recall value?**

Class 0: 0.92, out of 85, samples, 92 % were correctly identified as not purchased

Class 1: 0.88, out of 49 samples, 78 % were correctly identified as purchased

#### **6. whats the F1-score value?**

Class 0: Excellent balance between Recall and precision (0.90)

Class 1: Good balance between Recall and precision (0.81)

#### **7. whats the overall classification report?**

"The model achieved an overall accuracy of 87%, meaning it correctly classified 87% of the test data. For Class 0 (Not Purchased), the precision is 0.88%, recall is 92%, and F1-score is 90%, showing excellent performance. For Class 1 (Purchased), the precision is 84%, recall is 78%, and F1-score is 81%, indicating good performance, although the model misses some customers who actually purchased. Overall, the weighted F1-score is 85.5%, which shows the model performs well across both classes.

### **Question for Support Vector classification**

#### **1. How many class present in dataset?**

Dataset has 2 class :

Class 0 – Not purchased - 85

Class 1 – purchased - 49

#### **2. Does dataset is balanced?**

No dataset is imbalanced, because class 0 has 85 samples, class 1 has 49 samples

#### **3. Whats the overall performance of model?**

Accuracy – 78% precision – 0.79, Recall – 0.72, and F1 score = 0.73

#### **4. what the precision?**

class 0: 0.88, out of 85 samples, 76 % were identified as not purchased (irrespective of correct and incorrect)

class 1: 0.84, out of 49 samples, 88 % were identified as purchased irrespective of correct and incorrect)

#### **5. whats the recall value?**

Class 0: 0.92, out of 85, samples, 96% were correctly identified as not purchased

Class 1: 0.88, out of 49 samples, 47 % were correctly identified as purchased

## **6. whats the F1-score value?**

Class 0: Good balance between Recall and precision (0.85)

Class 1: poor balance between Recall and precision (0.61)

## **7. whats the overall classification report?**

"The model achieved an overall accuracy of 78%, meaning it correctly classified 78% of the test data. For Class 0 (Not Purchased), the precision is 0.76%, recall is 96%, and F1-score is 85%, showing good performance. For Class 1 (Purchased), the precision is 88%, recall is 47%, and F1-score is 61%, indicating poor performance, although the model misses some customers who actually purchased. Overall, the weighted F1-score is 73%, which shows the model doesn't performs well across both classes.

